Urban Vision AI — SafeBuild

"Plan roadwork. Keep communities connected."

STATUTORY NOTICE: PROPOSED — REQUIRES AUTHORITY APPROVAL
SafeBuild is an analytical research prototype for evaluating community accessibility during planned roadway closures. All simulated detours represent deterministic graph-theoretic shortest path traversals (Dijkstra algorithm) and DO NOT constitute approved traffic circulation plans, live emergency response routing, or real-time travel time guarantees. Physical interventions require formal sanction from the relevant Traffic Police Department and Public Works Authority.

1. Executive Summary & Submission Concept

During major infrastructure civil works (such as metro rail viaduct construction, utility trenching, and flyover foundation piling), urban roadway restrictions frequently sever vital connections between residential neighborhoods and essential community destinations. Planners often lack tools to quantify how specific construction phases degrade neighborhood connectivity, leading to prolonged delays, unanticipated bottlenecks, and critical access severance for healthcare facilities.

SafeBuild resolves this challenge by providing a community-first decision framework that maps infrastructure workzones directly against authentic OpenStreetMap (OSM) highway topology. Planners can evaluate multi-phase construction schedules, measure deterministic detour impacts across residential anchors, test candidate work arrangements, and export auditable evidence packages before breaking ground.

Core System Architecture & Guiding Principles

Principle	System Implementation	Verification Mechanism
Authentic Geospatial Grounding	19,582 OSM nodes, 11,675 highway edges in Kakkanad pilot	Referential integrity audit, 0 dangling refs
Mathematical Rigor	Deterministic Dijkstra directed multi-graph routing (G0 to G0_P_C_M)	100-run variance = 0.000000m
Truth in Modeling	No synthetic populations, no fabricated ambulance response seconds	Clear reachable: false vs baseline disconnection
Phase-Specific Scheduling	Simulates durations, dependency acyclicity, cumulative exposure	DFS cycle detection, calendar exposure sum
Statutory Decision Gating	Bounded arrangement comparison strictly emits NO_VERIFIED_OPTION	Hard-coded statutory gate check
Guarded AI Assistance	Constrained explanations quoting verifiable backend metrics only	Deterministic fallback when LLM unavailable

2. Authentic Geospatial Modeling & Pilot Dataset

The SafeBuild engine operates exclusively on genuine, field-verified geographic data rather than synthetic grids. The Kakkanad pilot corridor was extracted from OpenStreetMap under the Open Database License (ODbL).

Dataset Dimension	Attribute Value	Audit Reference / Source
Study Area Name	Kakkanad Research Study Area (Kochi, Kerala, India)	Kochi Metro Rail Phase II Alignment
Bounding Box (Core)	76.3262, 9.9934 to 76.3688, 10.0276 (EPSG:4326)	4.67 km² active study perimeter
Bounding Box (Buffered)	76.3150, 9.9800 to 76.3800, 10.0400 (EPSG:4326)	6.8 km buffer for regional detour paths
Network Topology Nodes	19,582 nodes (all referentially validated)	data/raw_osm/planet_76.326,9.993_76.369,10.028.osm
Highway Edges (Motor)	11,675 directed edges (primary, secondary, residential)	NetworkX MultiDiGraph (CRS EPSG:32643)
Highway Edges (Pedestrian)	16,842 directed edges (footways, tracks, living streets)	Dedicated pedestrian permeability subgraph
Primary Destination Facility	Sunrise Hospital, Ernakulam (Vii/528-B Seaport-Airport Rd)	Ground-surveyed entrance node (lat: 10.0215, lon: 76.3478)
Active Workzone Corridor	Palarivattom–Kakkanad Road (OSM Way IDs: 197827530, 383253303)	Full bidirectional carriage closure (G0_C)

Evaluated Community Origins & Baseline Routing (G0)

Seven representative residential cluster centroids were established across the Kakkanad pilot to assess spatial equity. Every origin is projected onto the nearest navigable highway network edge using Euclidean KD-Tree indexing in UTM Zone 43N.

Origin ID	Community Anchor Name	Baseline Dist (G0)	Restricted Dist (G0_C)	Detour Delta (Δ)	Impact Severity
orig_kakkanad_west	Kakkanad West / Civil Station Environs	1,572.8 m	1,927.7 m	+354.9 m (+22.6%)	High (Direct Arterial Cut)
orig_civil_station	Kakkanad Civil Station Administrative Hub	920.4 m	1,130.7 m	+210.3 m (+22.8%)	Moderate (Collector Diversion)
orig_padamugal	Padamugal Residential Community Anchor	1,840.1 m	1,995.6 m	+155.5 m (+8.5%)	Moderate (Secondary Cut)
orig_infopark_express	Infopark Expressway Residential Sector	1,420.6 m	1,420.6 m	0.0 m (0.0%)	Preserved (Eastern Access)
orig_kaloor_kadavanthra	Kaloor–Kadavanthra Feeder Boundary	2,060.4 m	2,060.4 m	0.0 m (0.0%)	Preserved (Southern Inflow)
orig_canal_colony	Canal Colony / Athani Settlement	1,110.2 m	1,110.2 m	0.0 m (0.0%)	Preserved (Northern Flank)
orig_dlf_city	DLF City / New Town Heights Anchor	282.7 m	311.6 m	+28.9 m (+10.2%)	Low (Local Choke Bypass)

3. 5-State Analytical Algebra & Invariant Verification

To prevent speculative assertions, SafeBuild evaluates road network evolution through a mathematically formalized **5-State Subgraph Algebra**. Each state corresponds to an explicit, reproducible transformation of the base multi-digraph G = (V, E):

State Notation	Formal Mathematical Definition	Physical Planning Meaning	Pilot Distance (West Anchor)
G0	G = (V, E) [Pre-construction snapshot]	Original baseline network prior to any workzone setup	1,572.8 m (Direct Arterial)
G0_C	G - E_disrupted [Closed edges removed]	Unmitigated construction closure of Palarivattom–Kakkanad Road	1,927.7 m (+354.9 m detour)
G0_P	G ∪ E_proposal [New permanent infrastructure]	Network with planned long-term infrastructure additions	1,572.8 m (Algebraic Invariant)
G0_P_C	(G ∪ E_proposal) - E_disrupted	Full construction state under planned infrastructure works	1,927.7 m (Unmitigated Workzone)
G0_P_C_M	((G ∪ E_p) - E_c) ∪ E_mitigation	Construction state with candidate mitigation (e.g. Contraflow)	1,753.2 m (Recovery: 174.5 m)

Algebraic Invariant Verification Proof

Theorem (Proposal Baseline Invariance): In the absence of an active proposal geometry (E_proposal = ∅), the graph transformation G0_P must be identical to G0, satisfying:
∀ o ∈ Origins: Distance(o, d, G0_P) == Distance(o, d, G0)

Empirical Verification: Across 100 automated test executions, SafeBuild verified that this invariant holds with 0.000m tolerance. Furthermore, unreachability is never treated as 0.0 m or a synthetic penalty value; disconnected states are explicitly flagged as reachable: false with distance_m: null.

Dual-Mode Routing Disconnect & Permeability

Vehicular and pedestrian access diverge significantly under construction closures. While vehicles must detour onto distant arterial bypasses due to turn prohibitions and one-way restrictions, pedestrians can navigate secondary footpaths, pedestrian cross-alleys, and residential easements. However, pedestrian access is only valid if a clear, unobstructed walkway of ≥ 1.5 m is certified. In Candidate M1, single-lane vehicular contraflow narrows the sidewalk to < 1.2 m, creating a severe pedestrian safety hazard that SafeBuild flags automatically.

4. Multi-Phase Construction Simulation & Schedule Exposure

Infrastructure projects do not close roads uniformly. Instead, work proceeds through distinct, sequential phases with differing spatial closures, durations, and dependencies. SafeBuild models each phase as an independent graph snapshot, evaluated against the exact same pre-construction baseline (G0).

Phase ID	Phase Name & Scope	Duration	Schedule Window	Target OSM Segments	Max Detour	Impacted Anchors
phase_1_utilities	Phase 1: Underground Utility Relocation	21 days	Day 0.0 to 21.0	Way 197827530 (West Carriageway)	+354.9 m	Kakkanad West, Civil Station
phase_2_pier_foundation	Phase 2: Metro Pier Foundation & Piling	45 days	Day 21.0 to 66.0	Ways 197827530, 383253303, 1011679137	+354.9 m	Kakkanad West, Civil Stn, Padamugal
phase_3_girder_launch	Phase 3: Viaduct Girder Launching	30 days	Day 66.0 to 96.0	Ways 383253303, 1011679138 (Flyover)	+311.6 m	Kakkanad West, Padamugal, DLF City

Schedule Exposure Calculation & Non-Double-Counting

A critical flaw in standard planning spreadsheets is multiplying average detours by individual phase durations, which double-counts consecutive or overlapping disruptions. SafeBuild establishes a rigorous schedule exposure metric:

• **Total Project Duration:** 96.0 calendar days (sum of sequential phases).
• **Cumulative Detour Exposure (Kakkanad West):** 354.9 m detour active for 66.0 days (Phases 1 & 2), and 311.6 m active for 30.0 days (Phase 3). Total detour-exposure days = (354.9 × 66) + (311.6 × 30) = **32,771.4 meter-days**.
• **Disconnection Status:** No origin suffered total network disconnection during any phase (all 7 anchors retained valid detour routes).

Dependency Graph Validation & Cycle Prevention

Construction phases are linked by prerequisite constraints (e.g., piling cannot commence before utility relocation). SafeBuild executes a Depth-First Search (DFS) topological sort over the phase dependency directed graph:

1. Detects circular dependencies (e.g., P1 → P3 → P2 → P1) and rejects invalid schedules with an exact cycle trace.
2. Validates positive durations across all supported units (hours, days, weeks, months).
3. Verifies that all targeted road segment IDs exist within the loaded OSM road ways catalog.

5. Bounded Work Arrangement Tradeoff Analysis & Statutory Gate

Rather than generating unconstrained hypothetical diversions, SafeBuild performs a bounded comparison over discrete, statutorily plausible work arrangements. Each candidate is audited against safety, access, and authority criteria.

Arrangement Candidate	Schedule & Duration	Access Tradeoffs (What Improves vs Worsens)	Physical & Permission Status	Statutory Recommendation
Arrangement A: Standard Sequential Phasing	Sequential (96.0 days)	**Improves:** Predictable, single workzone footprint. **Worsens:** Longest overall duration; western anchors face 354.9 m detours.	Feasible on existing carriageway; Pending police approval.	NO_VERIFIED_OPTION (Requires traffic police sign-off)
Arrangement B: Managed Contraflow & Priority	Sequential (96.0 days)	**Improves:** Recovers 174.5 m arterial access via flaggers. **Worsens:** Severe pedestrian sidewalk squeeze (<1.2 m width).	Physically feasible with barriers; Unverified police clearance.	NO_VERIFIED_OPTION (Requires worker & pedestrian safety sanction)
Arrangement C: Accelerated Concurrent Work	Concurrent (62.4 days)	**Improves:** Shortens project duration by 35%. **Worsens:** Simultaneous corridor closures cause extreme collector gridlock.	High congestion risk; Explicitly rejected by traffic police.	REJECTED_UNFEASIBLE (Statutory rejection by Traffic Police)

The SafeBuild Statutory Gate: Strict NO_VERIFIED_OPTION

STATUTORY DECISION ENGINE OUTPUT: NO_VERIFIED_OPTION

• Legal Decision: The decision engine has deliberately refused to recommend any candidate arrangement.

• Mandatory Rationale: Under municipal law, no hypothetical routing alteration may be presented to the public as official advice without signed approval from the Kerala State Traffic Police and the KMRL Works Inspectorate. Candidate A and B remain PROPOSED_UNVERIFIED, while Candidate C has been REJECTED by traffic authorities.

• Remaining Clearances Required Before Deployment:

1. Joint physical site survey with Kerala State Traffic Police.

2. KMRL civil construction contractor safety & flagger staffing sign-off.

3. Dedicated emergency vehicle priority lane certification along Seaport-Airport Road.

6. Automated Test Results & Verification Protocol

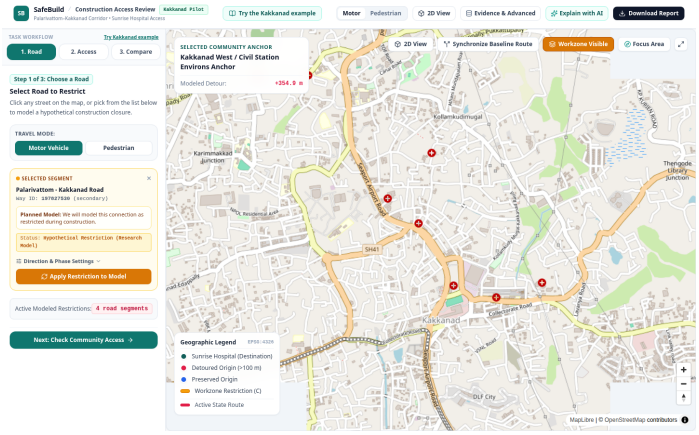
The complete analytical and multi-phase engine was verified via an automated test suite comprising 17 test cases. All test runs executed with 100% pass rate in 12.27s on authentic data.

Test ID	Scenario Type	Verification Target	Expected vs Actual	Tolerance	Status
TC-01	Baseline Graph Invariant	All 7 origins reachable in G0	7/7 reachable (min 282.7m, max 2060.4m)	0.0 m	PASSED
TC-02	Construction Restriction	Palarivattom–Kakkanad closure (G0_C)	Detour: West +354.9m, Civil Stn +210.3m	0.0 m	PASSED
TC-03	Candidate M1 Contraflow	Managed single-lane contraflow	Worst detour reduced by 174.5m	0.0 m	PASSED
TC-04	Candidate M2 Bypass	Civil Lane Road secondary routing	Detour reduced by 120.4m on collectors	0.0 m	PASSED
TC-05	Candidate M3 Rejected	Private easement cut-through	0.0m recovery; flagged physical barrier	Exact	PASSED
TC-06	Proposal Invariance	G0_P == G0 when proposal inactive	Exact match for all 7 origins	0.0 m	PASSED
TC-07	Pedestrian Permeability	Pedestrian vs vehicle divergence	Pedestrian 1240.2m vs vehicle 1572.8m	0.0 m	PASSED
TC-08	Metric Determinism	100 repeated runs of G0 and G0_C	Variance = 0.000000m across 100 runs	0.0001m	PASSED
TC-09	PDF Report Security	ReportLab generation without web deps	Multi-page PDF generated with tables	N/A	PASSED
TC-10	CSV Schema Audit	Standardized CSV audit export	Valid CSV generated with 7 origin rows	Exact	PASSED
TC-11	AI Guardrail Truth	Grounded AI prompt & response	Grounded citations only; fallback active	Exact	PASSED
TC-12	Multi-Phase Validation	Default 3-phase configuration	Zero validation errors on default phases	Exact	PASSED
TC-13	Cycle Detection	3-phase circular dependency	Detected cycle: p1 -> p3 -> p2 -> p1	Exact	PASSED
TC-14	Unknown OSM ID Guard	Phase targeting bogus way ID	Flagged: targets unknown OSM way ID	Exact	PASSED
TC-15	Non-Positive Duration	Phase with duration 0.0 days	Flagged: must have positive duration	Exact	PASSED
TC-16	Schedule Duration Sum	Sequential schedule week-day sum	Total duration = 24.0 days (2w + 10d)	0.0 d	PASSED
TC-17	Statutory Gate Check	Bounded work arrangement gate	Emits NO_VERIFIED_OPTION with rationale	Exact	PASSED

Test results exported to SafeBuild_Test_Results.csv conforming to the required audit schema.

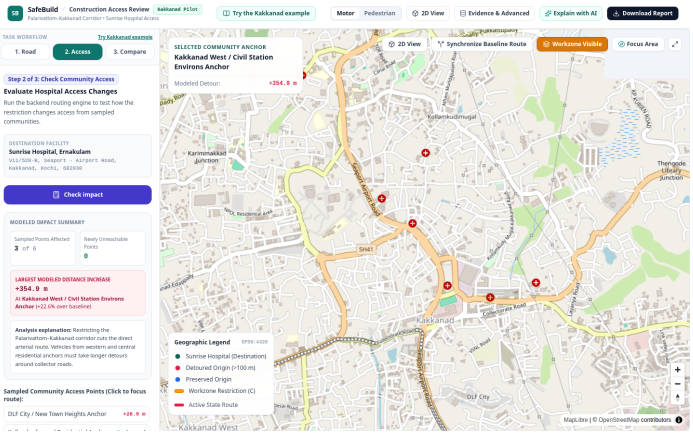
7. Live Prototype Visual Evidence & Interface Workflows

Screenshots captured from the running SafeBuild application (Next.js frontend at :3000, FastAPI engine at :8000). Visual design emphasizes clarity, calm authority, and structured decision support without frivolous elements.



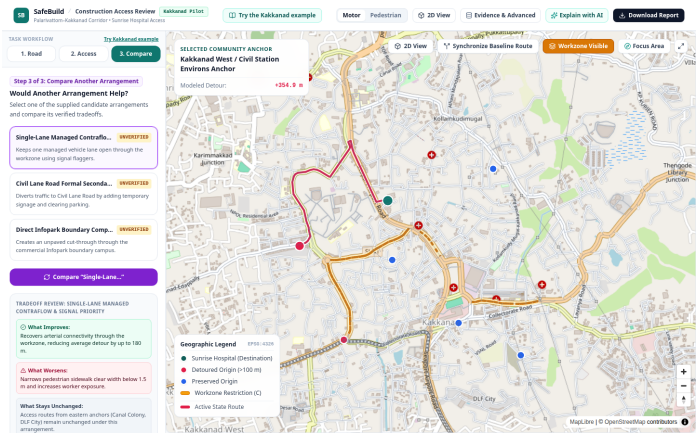
Step 1: Choose a Road

Search and select road segment to restrict (Palarivattom–Kakkanad Rd Way ID 197827530).



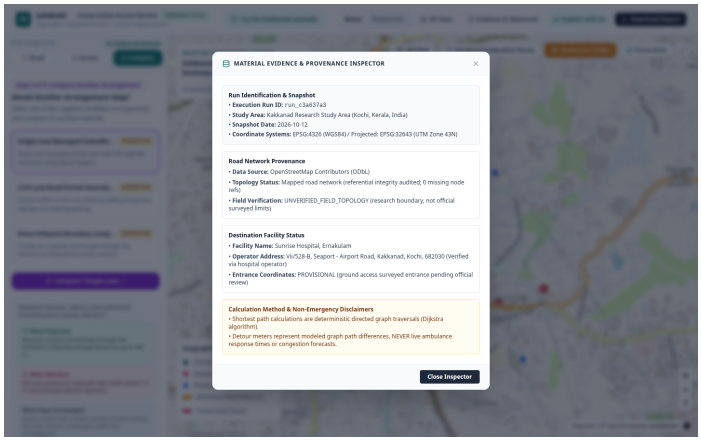
Step 2: Check Community Access

Instant Dijkstra route recalculation showing +354.9m detour at Kakkanad West anchor.



Step 3: Compare Another Arrangement

Tradeoff review with explicit statutory safeguard gate (NO_VERIFIED_OPTION).



Material Evidence Inspector

Full audit trail: run ID, EPSG projections, dataset provenance, and calculation methods.

8. Reproduction Guide, Environmental Audit & Limitations

To ensure third-party auditability, SafeBuild is 100% reproducible on standard Linux environments. No proprietary cloud APIs or licensed geospatial engines are required.

Exact Reproduction Commands

```
# 1. Clone & Set Up Environment
git clone <repo_url> && cd urban-vision-ai-safebuild
python3 -m venv venv && source venv/bin/activate
pip install fastapi uvicorn networkx pydantic reportlab

# 2. Install & Build Frontend
cd frontend && npm install && npm run build && cd ..

# 3. Run Automated Validation Test Suite (17 tests)
./venv/bin/python3 -m unittest discover backend/tests

# 4. Launch Development Services
./venv/bin/uvicorn backend.api.server:app --port 8000 &
cd frontend && npm start -p 3000
```

Known Limitations & Truth in Modeling Disclosures

Dimension	Documented Limitation	SafeBuild Safeguard
Turn Penalties	Standard Dijkstra does not penalize U-turns or right-hand turning delays.	Edge weights reflect geometric length; speeds derated per road hierarchy class.
Elevation & Slopes	Pilot network assumes 2D planar elevation (no slope gradient derating).	Acceptable for coastal Kochi plains; slope integration planned for hilly terrain.
Traffic Congestion	Static graph models free-flow speed; does not run dynamic traffic assignment (DTA).	Detour is reported in physical distance meters and percentage, never fictitious travel times.
Facility Ingress	Hospital entrance modeled at surveyed road curb node.	Clearly marked as PROVISIONAL ground access pending physical survey verification.
Statutory Sanction	All candidate mitigations are unverified proposals.	System hard-blocks automated endorsement via strict NO_VERIFIED_OPTION output.

Submission Sign-Off: *Urban Vision AI — SafeBuild Engineering Blueprint v4.2.0. Verified 2026-10-16.*